

Add Functions to Scripts

MATLAB® scripts, including live scripts, can contain code to define functions. These functions are called *local functions*. Local functions are useful if you want to reuse code within a script. By adding local functions, you can avoid creating and managing separate function files. They are also useful for experimenting with functions, which can be added, modified, and deleted easily as needed.

Create a Script with Local Functions

To create a script or live script with local functions, go to the **Home** tab and select **New Script** or **New Live Script**. Then, add code to the file. Each local function must begin with its own function definition statement and end with the `end` keyword. The functions can appear in any order and can be defined anywhere in the script.

Before R2024a: Local functions in scripts must be defined at the end of the file, after the last line of script code.

For example, create a script called `mystats.m`.

```
edit mystats
```

In the file, include two local functions, `mymean` and `mymedian`. The script `mystats` declares an array, determines the length of the array, and then uses the local functions `mymean` and `mymedian` to calculate the average and median of the array.

```
x = 1:10;
n = length(x);

function a = mymean(v,n)
% MYMEAN Local function that calculates mean of array.

    a = sum(v)/n;
end

function m = mymedian(v,n)
% MYMEDIAN Local function that calculates median of array.

    w = sort(v);
    if rem(n,2) == 1
        m = w((n + 1)/2);
    else
        m = (w(n/2) + w(n/2 + 1))/2;
    end
end

avg = mymean(x,n);
med = mymedian(x,n);
```

Run Scripts with Local Functions

To run a script or live script that includes local functions, in the **Editor** or **Live Editor** tab, click the  **Run** button. You also can type the saved script name in the Command Window.

To run an individual section inside a script or live script, place the cursor inside the section and use the  **Run Section** button (requires R2017b or later for `.m` files). In live scripts or functions, you only can run sections that are before the local function definitions.

Restrictions for Local Functions and Variables

Local functions are only visible within the file where they are defined. They are not visible to functions in other files, and cannot be called from the Command Window.

Local functions in the current file have precedence over functions in other files. That is, when you call a function within a script, MATLAB checks whether the function is a local function before looking for the function in other locations. This allows you to create an alternate version of a particular function while retaining the original in another file.

Scripts create and access variables in the base workspace. Local functions, like all other functions, have their own workspaces that are separate from the base workspace. Local functions cannot access variables in the workspace of other functions or in the base workspace, unless you pass them as arguments.

Scripts that contain local functions must contain executable code outside of a function.

Access Help for Local Functions

Although you cannot call a local function from the command line or from functions in other files, you can access its help using the `help` command. Specify the names of both the script and the local function, separating them with a `>` character.

For example:

```
help mystats>mymean
```

```
mymean Local function that calculates mean of array.
```

See Also

Topics

[Create Functions in Files](#)

[Function Precedence Order](#)

[Base and Function Workspaces](#)